

December 2019

ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.
Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.

Answer any five questions from the rest.

1. Fill in the blanks (any twenty): 1 X 20
- i) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
 - ii) Grey cast iron is _____ than white cast iron.
 - iii) The iron ore mostly used for the production of pig iron is _____.
 - iv) The plasticity is that property of a material by virtue of which it deforms _____ under the action of external force.
 - v) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
 - vi) The ultimate stress of a material is _____ than the breaking stress.
 - vii) Cutting tools are made from _____ steel.
 - viii) Grey cast iron is _____ than white cast iron.
 - ix) The iron ore mostly used for the production of pig iron is _____.
 - x) Ball bearing are usually, made from _____ steel.
 - xi) An alloy of copper, tin and zinc is known as _____.
 - xii) German silver consists of _____.
 - xiii) Babbit metal is a _____ based alloy.
 - xiv) The metal suitable for bearing subjected to high load is _____.
 - xv) Brass is an alloy of _____ and _____.
 - xvi) The heat treatment process used for softening hardened steel is _____.
 - xvii) The process used for relieving the internal stresses previously set up in the metal and for increasing the machinability of steel is _____.
 - xviii) Carburizing is a _____ process.

- xix) Normalizing _____ the mechanical properties of steel.
- xx) Normalizing of steel is done to _____ the grain structure.
- xxi) Phenolic resins are properly known as _____.
- xxii) The property of a material by virtue of which it can withstand or support an external force or load without rupture, is known as _____.
2. a) What do you mean by heat treatment of steel?
b) Describe any two of the following heat treatment processes – i) cyaniding, ii) carburising, iii) induction hardening, iv) nitriding. 4+6
3. a) Write a short note on corrosion.
b) What is the difference between creep and fatigue?
c) Write a short note on – i) pitting, ii) erosion. 3+3+4
4. a) Define alloy steel. Explain the effect of nickel, chromium and manganese on the properties of steel.
b) Describe aluminium casting alloys and Y-alloy in brief. 6+4
5. Describe: blending, compacting, pre-sintering and sintering method. 10
6. a) Write briefly about the different stages of powder metallurgy.
b) State the advantages of powder metallurgy. 8+2
7. a) How do thermo plastics differ from thermosetting plastics?
b) Write short notes on – i) Thermocole, ii) Glass wool.
c) State the properties of refractory materials. 3+4+3
8. a) Draw iron carbon equilibrium diagram and write the name of different phases and mention different.
b) What is the Duralumine? 7+3
9. a) State the properties and applications of copper alloy.
b) What is bearing material? 7+3
10. a) Difference between destructive testing and non destructive testing.
b) Write a short note on – i) Dry penetrant test, ii) Magnaflux test. 5+5

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